Pinned group summary: `SU(n=4, q=2, eps=-1)`

Pinning-Sympy automated output

July 6, 2026

Contents

1 Basic group attributes

Group	`SU(n=4, q=2, eps=-1)`
Matrix size	4×4
Rank	2
Defining equations	$DefiningEquationPlaceholder$

1.1 Skew-hermitian form

Dimension	4
Witt index	2
Primitive element	$\sqrt{d}$
Epsilon	-1

Matrix: $\begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{bmatrix}$

1.2 Generic Lie algebra element

$$\begin{bmatrix} \sqrt{d}x_{i00} + x_{r00} & \sqrt{d}x_{i01} + x_{r01} & -x_{r02} & -\sqrt{d}x_{i12} + x_{r12} \\ \sqrt{d}x_{i10} + x_{r10} & -\sqrt{d}x_{i00} + x_{r11} & \sqrt{d}x_{i12} + x_{r12} & -x_{r13} \\ -x_{r20} & -\sqrt{d}x_{i30} + x_{r30} & \sqrt{d}x_{i00} - x_{r00} & \sqrt{d}x_{i10} - x_{r10} \\ \sqrt{d}x_{i30} + x_{r30} & -x_{r31} & \sqrt{d}x_{i01} - x_{r01} & -\sqrt{d}x_{i00} - x_{r11} \end{bmatrix}$$

1.3 Generic torus element

$$\begin{bmatrix} t_0 & 0 & 0 & 0 \\ 0 & t_1 & 0 & 0 \\ 0 & 0 & \frac{1}{t_0} & 0 \\ 0 & 0 & 0 & \frac{1}{t_1} \end{bmatrix}$$

1.4 Trivial characters

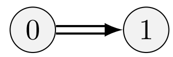
$$\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

2 Root system

Summary Properties

Dynkin type	`B2`
Irreducible	True
Reduced	True
Simply laced	False
Number of roots	8

2.1 Dynkin diagram



2.2 Cartan matrix

$$\begin{bmatrix} 2 & -2 \\ -1 & 2 \end{bmatrix}$$

2.3 Roots and coroots

Root	**Coroot**	**Simple**	**Sign**	**Multipliable**
$(-2,\ 0,\ 0,\ 0)$	$(-1,\ 0,\ 1,\ 0)$	No	$-$	No
$(-1,\ -1,\ 0,\ 0)$	$(-1,\ -1,\ 1,\ 1)$	No	$-$	No
$(-1,\ 1,\ 0,\ 0)$	$(-1,\ 1,\ 1,\ -1)$	No	$-$	No
$(0,\ -2,\ 0,\ 0)$	$(0,\ -1,\ 0,\ 1)$	No	$-$	No
$(0,\ 2,\ 0,\ 0)$	$(0,\ 1,\ 0,\ -1)$	Yes	$+$	No
$(1,\ -1,\ 0,\ 0)$	$(1,\ -1,\ -1,\ 1)$	Yes	$+$	No
$(1,\ 1,\ 0,\ 0)$	$(1,\ 1,\ -1,\ -1)$	No	$+$	No
$(2,\ 0,\ 0,\ 0)$	$(1,\ 0,\ -1,\ 0)$	No	$+$	No

2.4 Linear combinations of roots

Definition 2.1. For two roots α, β in a root system Φ, the set of positive integral linear combinations of α, β that are also roots is denoted $(\alpha, \beta) = \{i\alpha + j\beta \in \Phi : i, j \in \mathbb{Z}_{\geq 1}\}$

α	β	Pairs (i, j)
$(-2,\ 0,\ 0,\ 0)$	$(1,\ -1,\ 0,\ 0)$	$(1,1),(1,2)$
$(-2,\ 0,\ 0,\ 0)$	$(1,\ 1,\ 0,\ 0)$	$(1,1),(1,2)$
$(-1,\ -1,\ 0,\ 0)$	$(-1,\ 1,\ 0,\ 0)$	$(1,1)$
$(-1,\ -1,\ 0,\ 0)$	$(0,\ 2,\ 0,\ 0)$	$(1,1),(2,1)$
$(-1,\ -1,\ 0,\ 0)$	$(1,\ -1,\ 0,\ 0)$	$(1,1)$
$(-1,\ -1,\ 0,\ 0)$	$(2,\ 0,\ 0,\ 0)$	$(1,1),(2,1)$
$(-1,\ 1,\ 0,\ 0)$	$(0,\ -2,\ 0,\ 0)$	$(1,1),(2,1)$
$(-1,\ 1,\ 0,\ 0)$	$(1,\ 1,\ 0,\ 0)$	$(1,1)$
$(-1,\ 1,\ 0,\ 0)$	$(2,\ 0,\ 0,\ 0)$	$(1,1),(2,1)$
$(0,\ -2,\ 0,\ 0)$	$(1,\ 1,\ 0,\ 0)$	$(1,1),(1,2)$
$(0,\ 2,\ 0,\ 0)$	$(1,\ -1,\ 0,\ 0)$	$(1,1),(1,2)$
$(1,\ -1,\ 0,\ 0)$	$(1,\ 1,\ 0,\ 0)$	$(1,1)$

3 Root spaces

Root	Dimension
$(-2,\ 0,\ 0,\ 0)$	1
$(-1,\ -1,\ 0,\ 0)$	2
$(-1,\ 1,\ 0,\ 0)$	2
$(0,\ -2,\ 0,\ 0)$	1
$(0,\ 2,\ 0,\ 0)$	1
$(1,\ -1,\ 0,\ 0)$	2
$(1,\ 1,\ 0,\ 0)$	2
$(2,\ 0,\ 0,\ 0)$	1

RootSpaceTablePlaceholder

4 Root subgroups

RootSubgroupTablePlaceholder
HomomorphismDefectTablePlaceholder
HomomorphismDefectIdentitiesPlaceholder

5 Commutators

CommutatorCoefficientTablePlaceholder
CommutatorIdentitiesPlaceholder

6 Weyl group

WeylGroupPlaceholder
WeylConjugationCoefficientPlaceholder
WeylConjugationIdentitiesPlaceholder

7 Coroot torus elements (h_α)

CorootTorusElementPlaceholder